



**UNIVERSITÀ
DEGLI STUDI
DI UDINE**
hic sunt futura

DEPARTMENT OF MATHEMATICS, COMPUTER SCIENCE AND PHYSICS

Master Degree: Artificial Intelligence & Cybersecurity

A comparative analysis of Latent Architectures

Structural Optimization and Discriminative Power in High-Dimensional Data.

Damiano Fumagalli

ID: 157547

Contact: fumagalli.damiano@spes.uniud.it

2025/2026



Manifold Hypothesis: in high-dimensional spaces, data becomes sparse, and Euclidean distances become uninformative. This sparsity implies that the data occupies a lower-dimensional Manifold.

Goal of the presentation: show how different autoencoder architectures exploit this manifold for feature extraction and data compression

Case Study: using Fashion-MNIST dataset train different kind of autoencoders and evaluate the latent space fidelity with some metrics and classification tasks

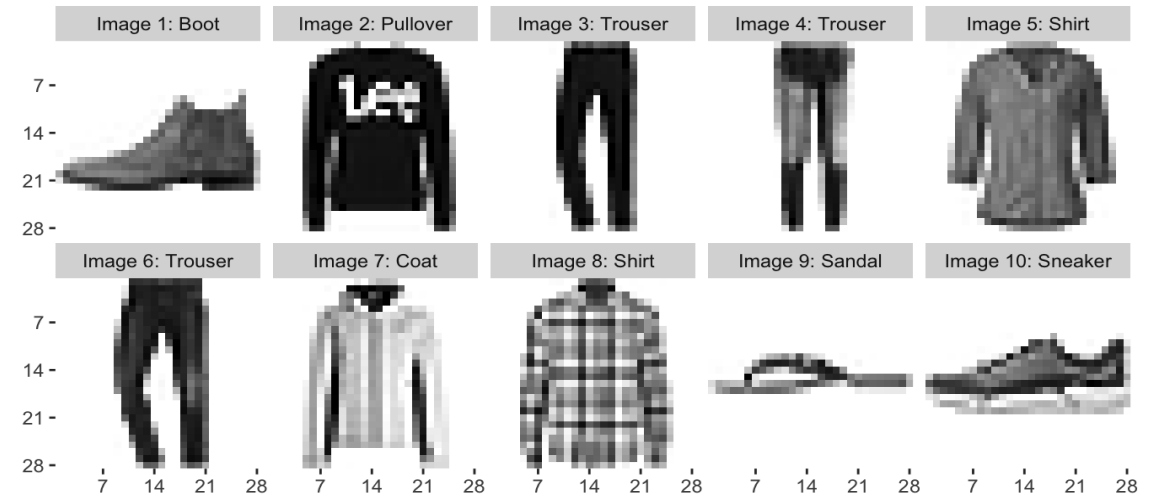


Figure 1: Samples from Fashion-MNIST dataset

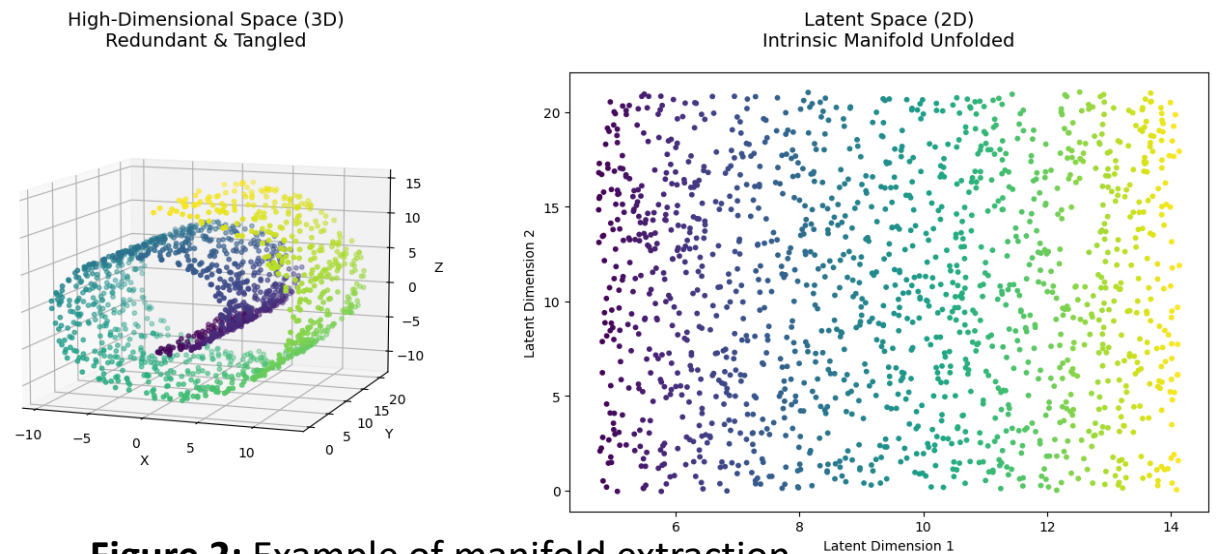
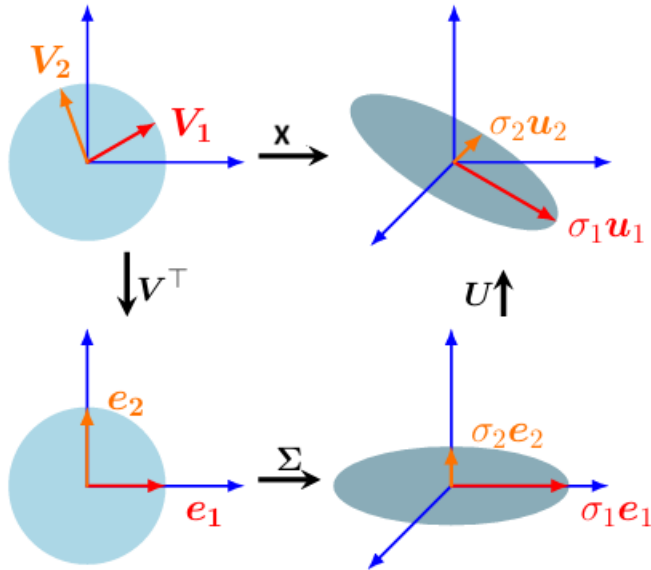
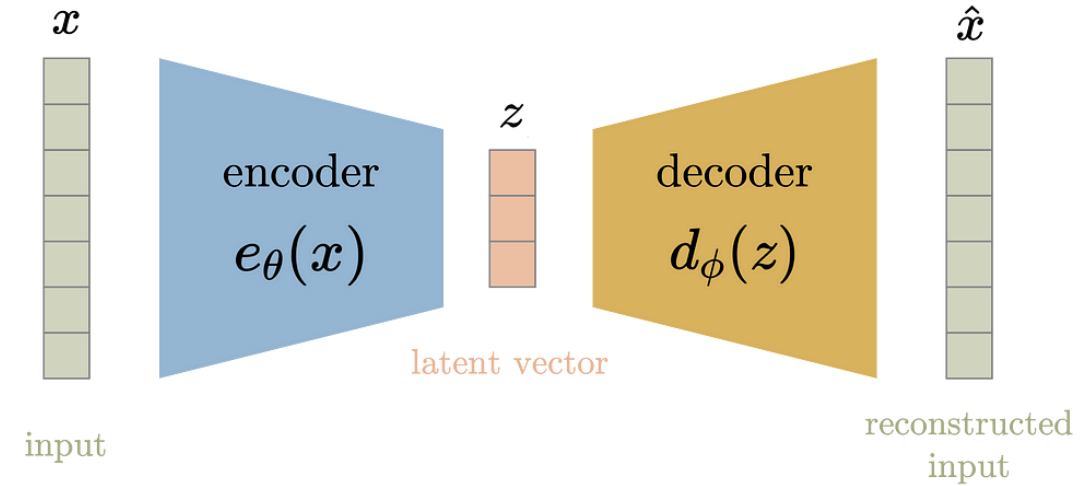


Figure 2: Example of manifold extraction



- $X \in \mathbb{R}^D \rightarrow Z \in \mathbb{R}^d, d \ll D$
- **PCA (k):** $Z = XV_k$
 - Goal: minimize the reconstruction loss

$$L = \left\| X - XV_k V_k^T \right\|_F^2$$
 - V_k is obtained by the first k components of the right singular values V of the SVD
 - **SVD:** $\hat{X} = U \Sigma V^T$

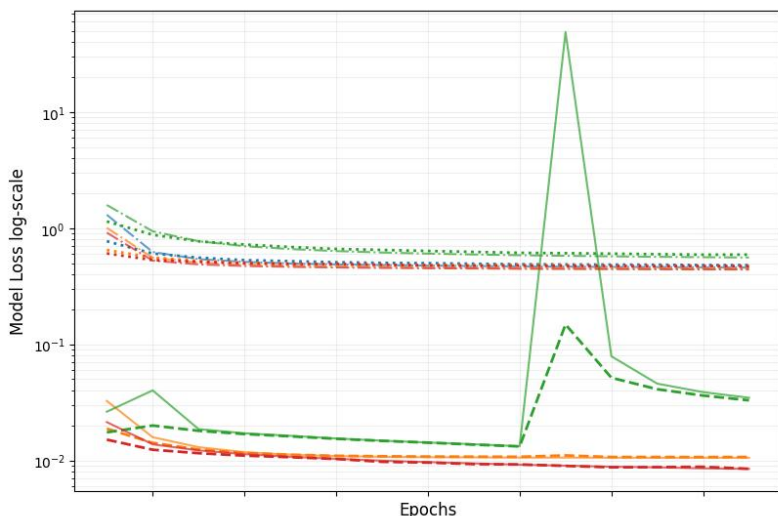


$$loss = \|x - \hat{x}\|_2 = \|x - d_\phi(z)\|_2 = \|x - d_\phi(e_\theta(x))\|_2$$

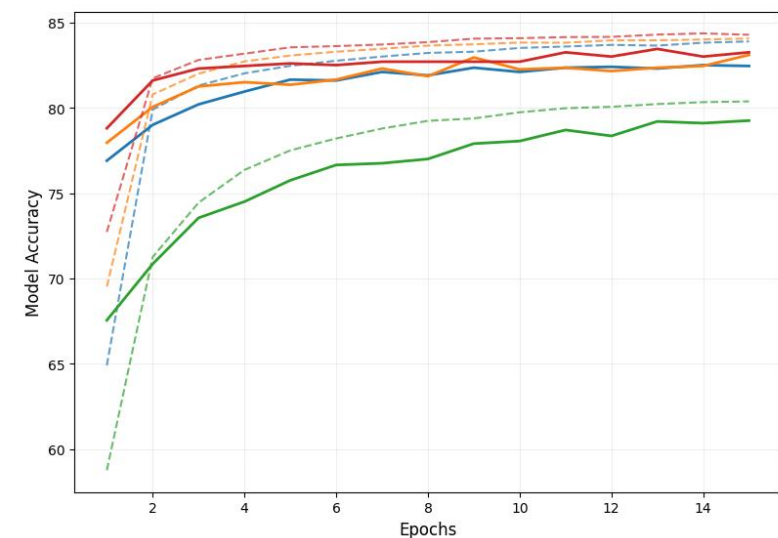
- $e_\theta(x): \mathbb{R}^D \rightarrow \mathbb{R}^d$ produces $Z = XW_\theta$
- $d_\phi: \mathbb{R}^d \rightarrow \mathbb{R}^D$ produces $\hat{X} = ZW_\phi = XW_\theta W_\phi$
- Goal: minimize reconstruction loss $L = \left\| X - XW_\theta W_\phi \right\|_F^2$



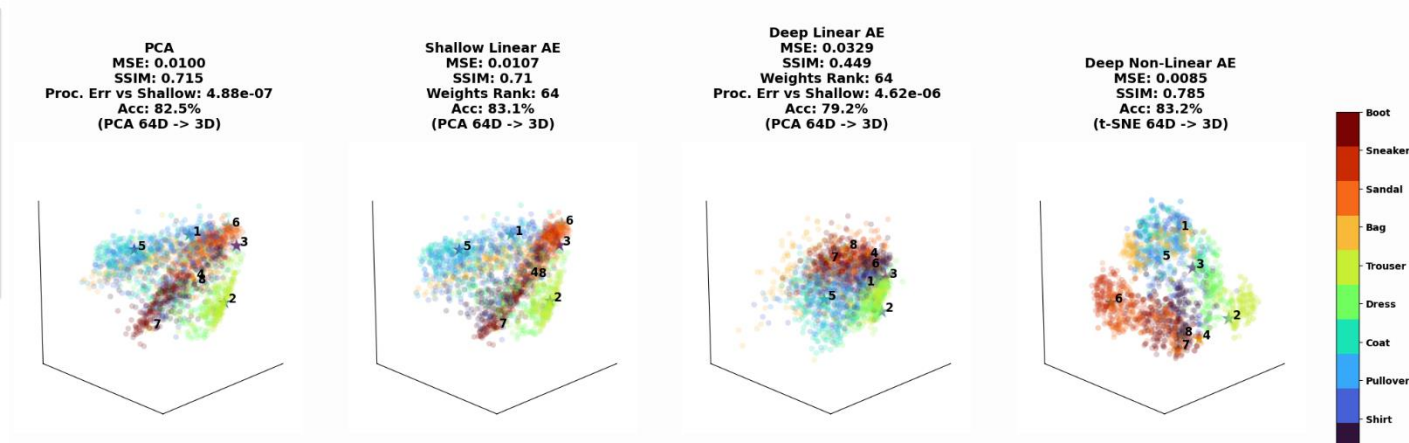
Training & Validation Loss



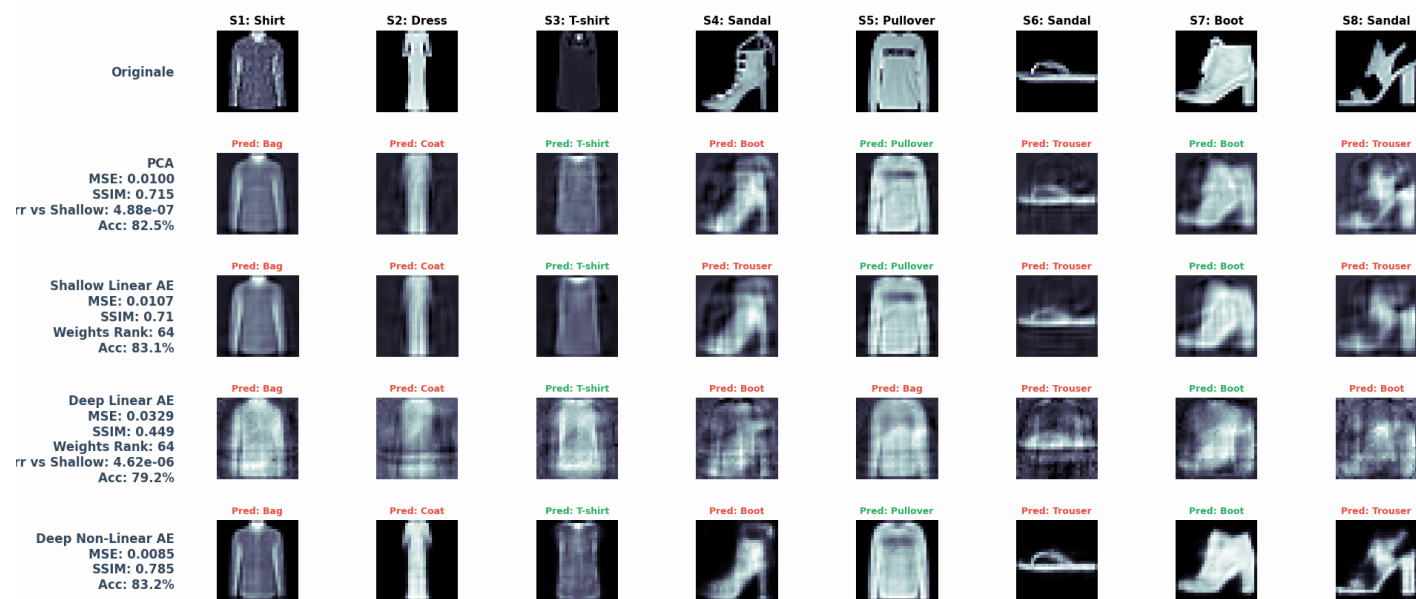
Training & Validation Accuracy

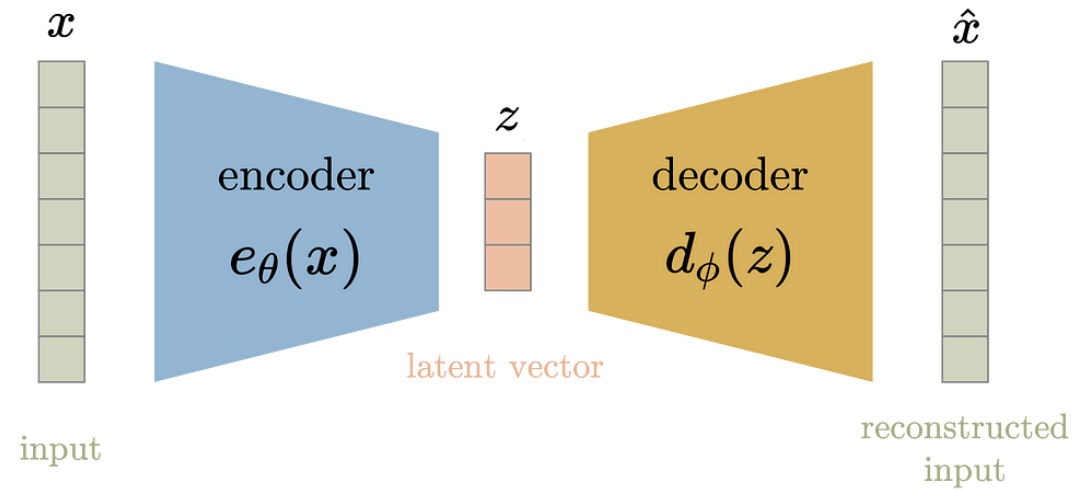


64D Latent Space Comparison

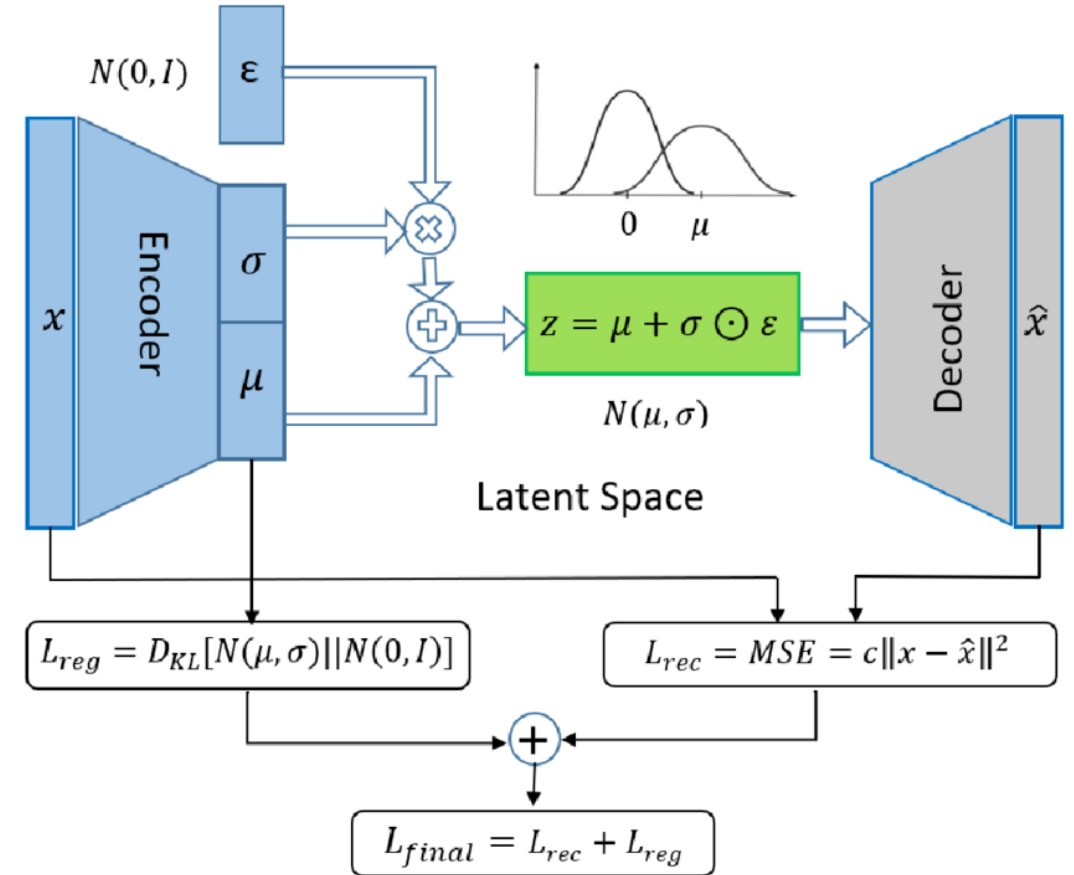


Sample Reconstruction Comparison



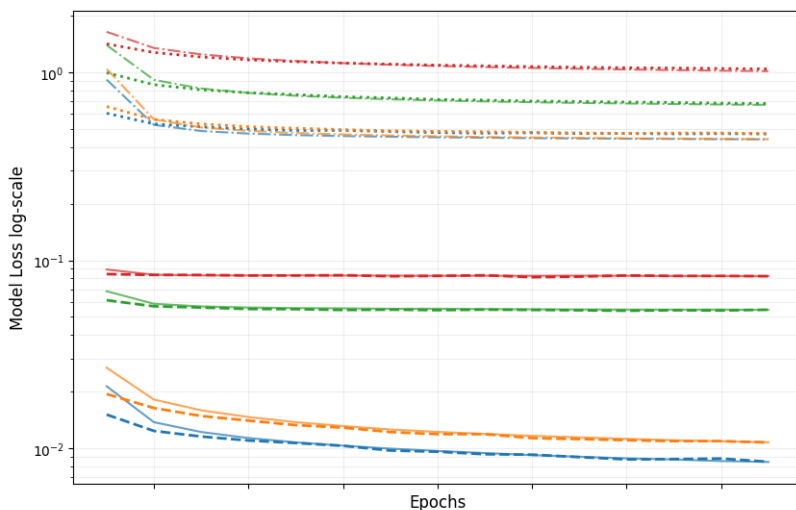


$$loss = \|x - \hat{x}\|_2 = \|x - d_\phi(z)\|_2 = \|x - d_\phi(e_\theta(x))\|_2$$

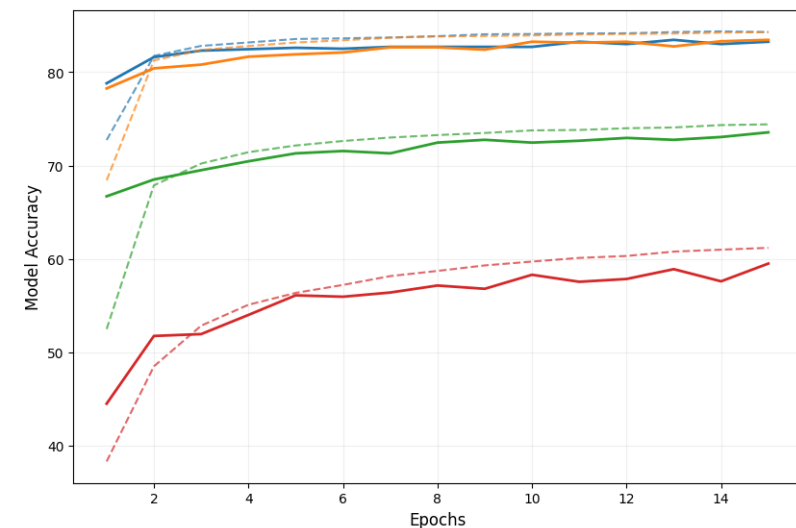




Training & Validation Loss

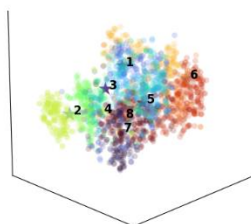


Training & Validation Accuracy

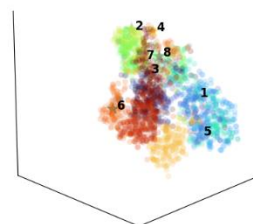


64D Latent Space Comparison

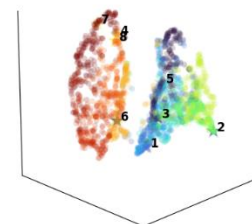
Standard AE
SSIM: 0.785
MIG: 0.0112
Silh: 0.0533
Acc: 83.2%
Proc. Err vs beta=0.01 VAE: 2.28e-06
(t-SNE 64D -> 3D)



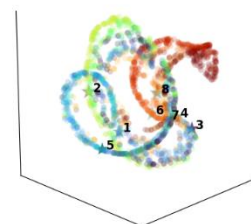
beta=0.01 VAE
SSIM=0.763
MIG: 0.111
Silh: 0.0448
Acc: 83.5%
(t-SNE 64D -> 3D)



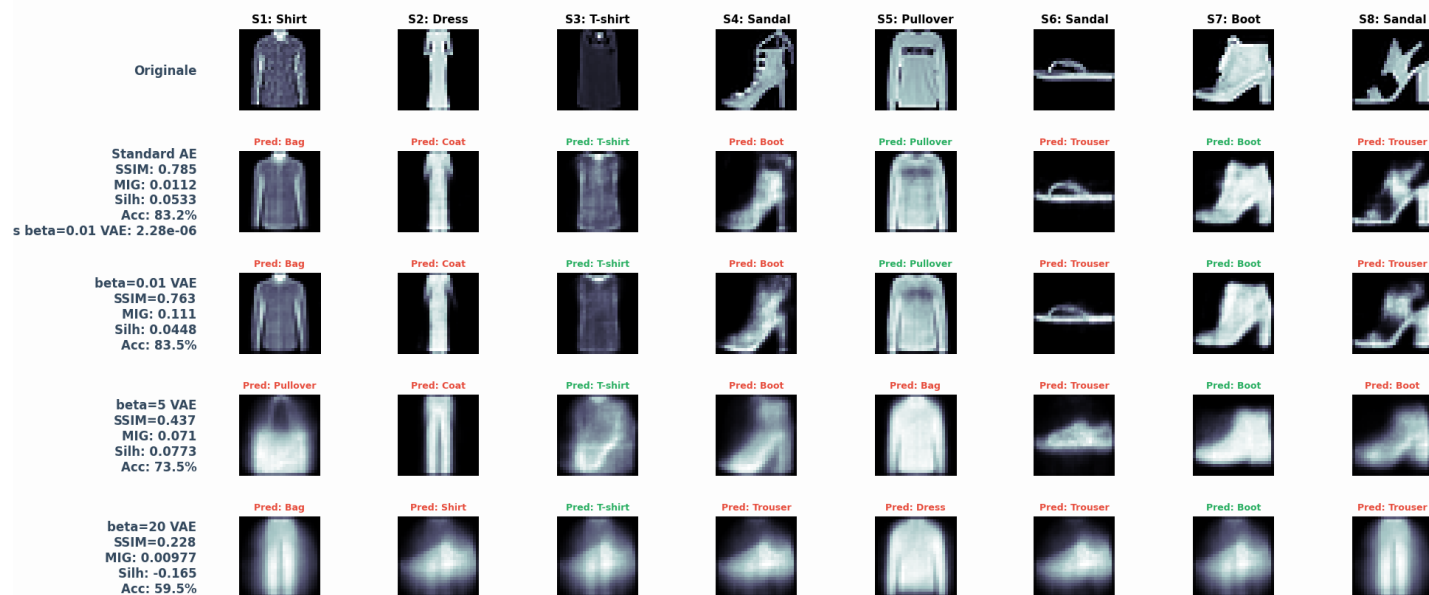
beta=5 VAE
SSIM=0.437
MIG: 0.071
Silh: 0.0773
Acc: 73.5%
(t-SNE 64D -> 3D)

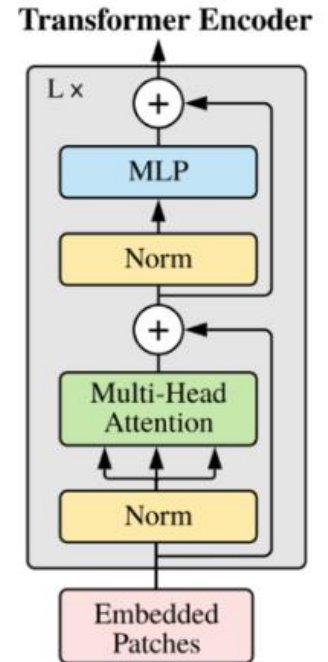
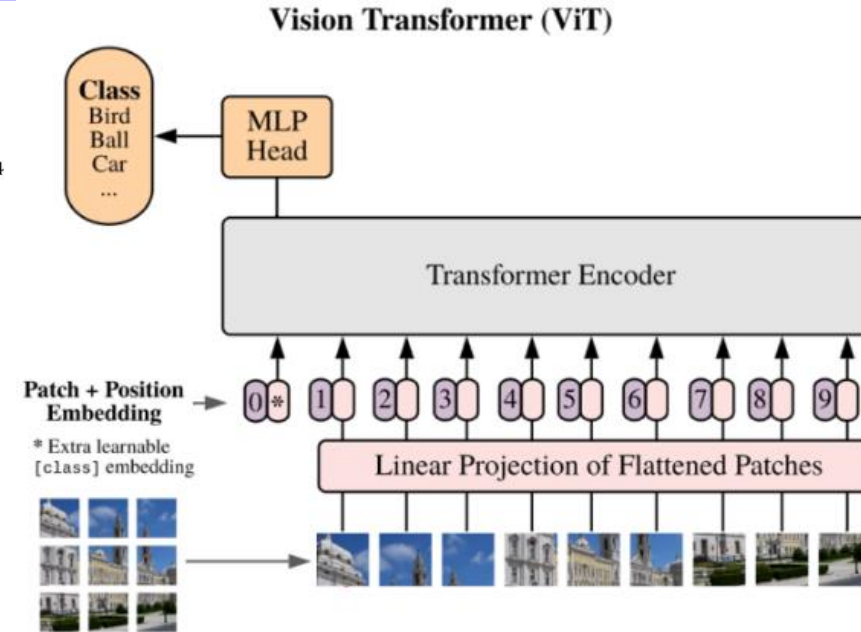
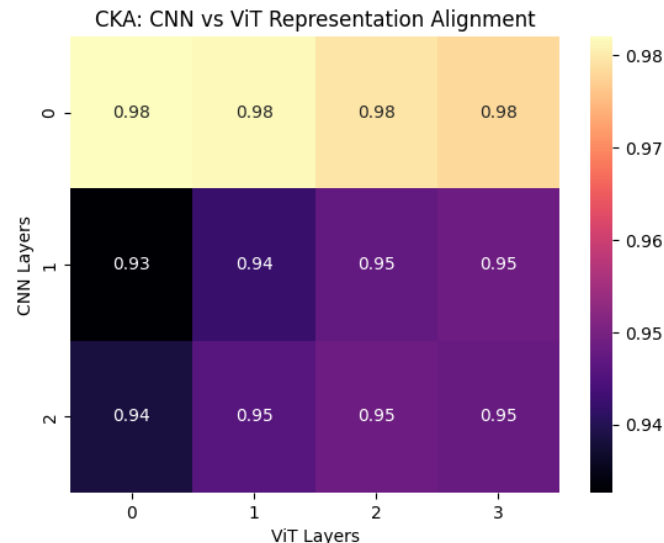
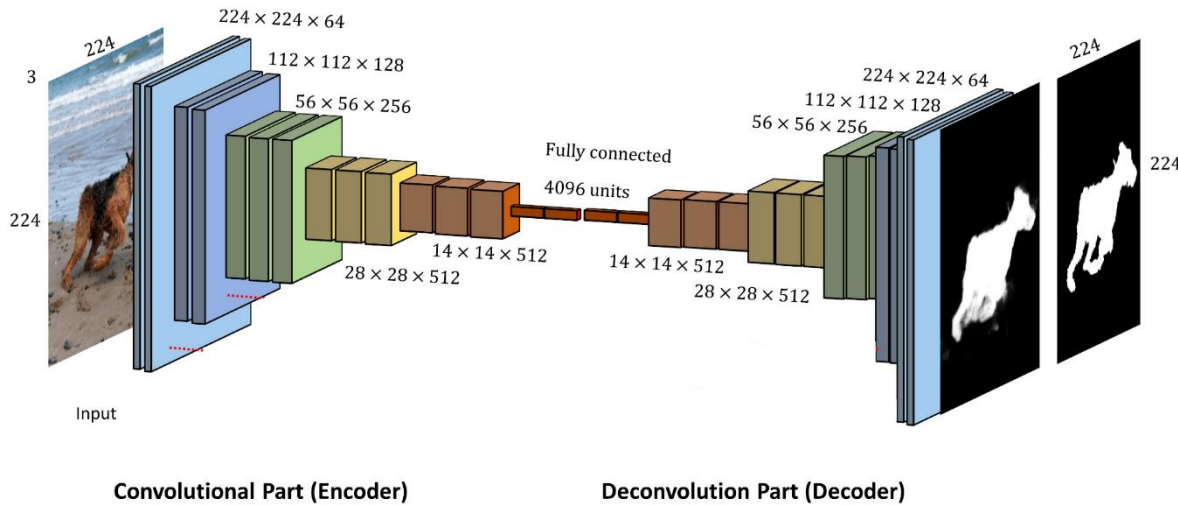


beta=20 VAE
SSIM=0.228
MIG: 0.00977
Silh: -0.165
Acc: 59.5%
(t-SNE 64D -> 3D)



Sample Reconstruction Comparison

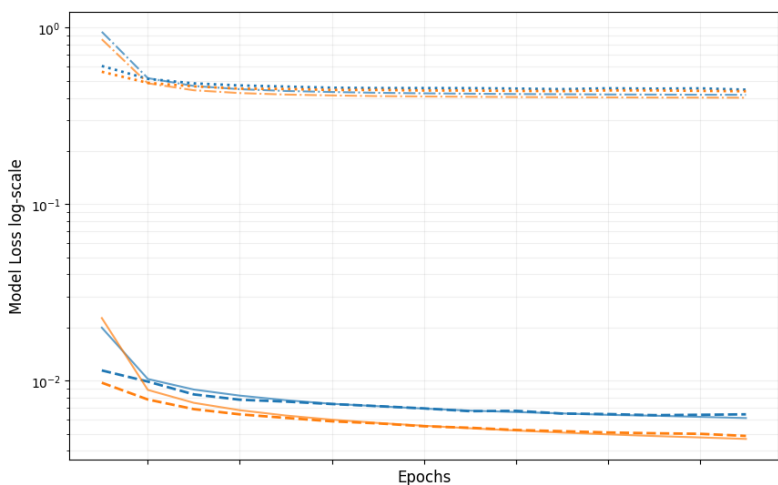




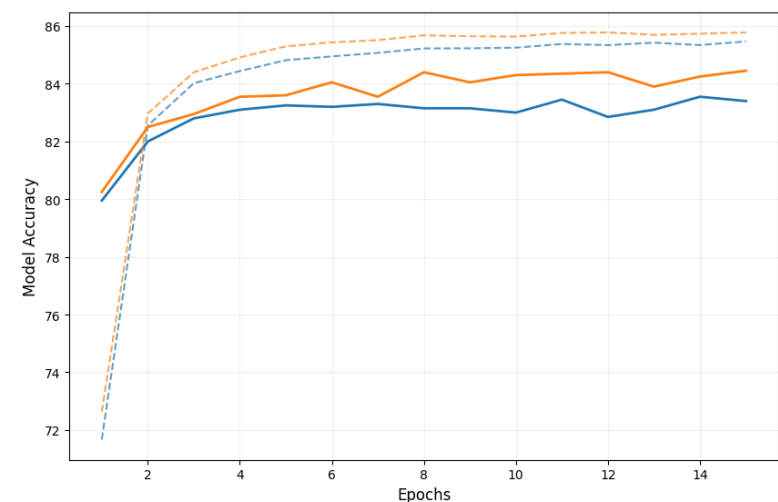
- [“Do Vision Transformers See Like Convolutional Neural Networks?”](#)
Maithra Raghu, Thomas Unterthiner
- [“On the Relationship between Self-Attention and Convolutional Layers”](#)
Jean-Baptiste Cordonnier, Andreas Loukas, Martin Jaggi



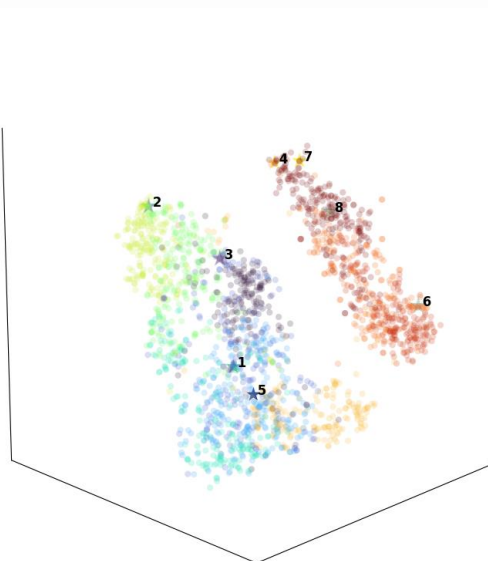
Training & Validation Loss



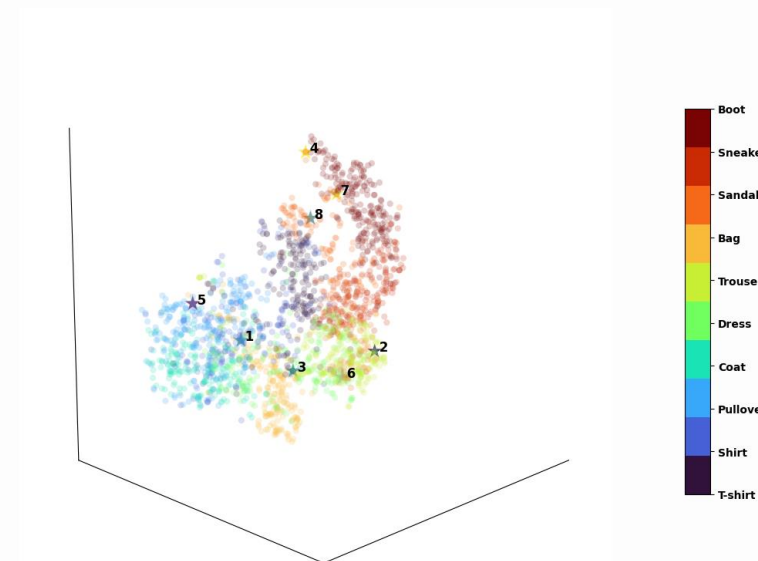
Training & Validation Accuracy



Conv AE
SSIM: 0.833
Silh: 0.0482
MegaFLOPs=2.78
Params count=8.58e+05
Acc: 83.4%
(t-SNE 64D -> 3D)



Transformer
SSIM: 0.873
Silh: 0.0628
MegaFLOPs=78.6
Params count=2.41e+06
Acc: 84.5%
(t-SNE 64D -> 3D)



64D Latent Space Comparison

Sample Reconstruction Comparison

